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Potamides: JAX tools for curvature-based inference from stellar streams

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Summary

`Potamides` is a Python package for inferring the mass distribution of galaxies from the projected shapes of stellar streams in imaging data. Stellar streams are elongated structures produced when star clusters or dwarf galaxies are tidally disrupted by their host. Because their projected tracks carry information about the host's gravitational field, the local curvature of a stream can constrain the underlying potential.

The package implements and extends the curvature-based likelihood framework of (Nibauer et al., 2023). Rather than generating a full dynamical realization of a stellar stream for each trial model, `Potamides` represents observed stream tracks with JAX-based splines. It evaluates gravitational accelerations in candidate potentials and compares them directly to the local stream geometry. This provides a lower-cost inference workflow that complements traditional forward-modeling approaches. `Potamides` supports the complete analysis pipeline for a galaxy, from annotating stream ridge-lines to evaluating likelihoods across many potential models.

Statement of need

Stellar streams are popular tracers of galactic gravitational potentials and the dark matter halos that dominate galaxies (Bonaca et al., 2014). For external galaxies, the observed dynamical information is often limited to projected stream morphology. The curvature-based method of (Nibauer et al., 2023) addresses this regime by using the local relationship between stream curvature and gravitational acceleration to directly constrain the potential's geometry from the projected stream track.

Until now, this method lacked a reusable, high-performance software implementation intended for community use. `Potamides` fills that gap by serving as the active platform for curvature-based inference. It builds on the original reference implementation while introducing methodological improvements such as spline-knot optimization, better handling of locally straight segments, and joint inference across multiple streams within a common likelihood.

The package is designed for researchers who need to analyze stream systems end-to-end. Users can estimate a smooth ridge-line from ordered points, extract curvature observables, and evaluate families of gravitational potentials efficiently enough to test a wide variety of model assumptions.

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State of the field

`Potamides` occupies a different methodological niche from commonly used galactic-dynamics packages such as `gala` (Price-Whelan, 2017), `galpy` (Bovy, 2015), and `AGAMA` (Vasiliev, 2019). Those libraries provide excellent tools for orbit integration, action-angle methods, and forward modeling. They are ideal when the goal is to simulate orbits or generate stream realizations from explicit physical histories. Recent JAX-based packages like `galax` (Starkman et al., 2024) and `StreamSculptor` (Nibauer et al., 2025) bring differentiable, GPU-compatible modeling to the ecosystem, but they similarly focus on classical dynamical calculations and forward simulations.

`Potamides` is designed to complement these tools. It is not a general galactic-dynamics library. Instead, it treats the projected stream track as the primary observable and compares it directly with the local acceleration field implied by a candidate potential. This approach makes fewer assumptions about progenitor properties and stream formation history than forward models. It is best suited for problems where fast, direct constraints from morphology are required, with forward modeling serving as a possible complementary follow-up.

Software design

The core design goal of `Potamides` is to make curvature-based inference a practical end-to-end workflow by coupling geometric track modeling, likelihood evaluation, and visualization within a single framework.

The JAX-first implementation (Bradbury et al., 2018) enables vectorization, just-in-time compilation, and hardware portability without requiring separate code paths for prototyping and production. `Potamides` uses `unxt` (Starkman et al., 2025) for unit-aware quantities and interoperates with `galax` (Starkman et al., 2024) for potential evaluation. This allows users to test flexible halo and disc models within a single compiled pipeline.

Stream tracks are represented with splines rather than raw point sets. `Potamides` provides utilities for constructing smooth ridge-line splines from ordered stream points, optimizing knot placement, and computing the resulting tangent vectors, principal normals, and scalar curvature. The package also includes specific handling for locally straight segments where naive curvature treatments might bias the likelihood.

Real stream systems are often fragmented or only partly observed. To address this, `Potamides` evaluates segment-wise likelihoods and combines them across multiple structures, treating curvature inference as a unified process over an entire galaxy.

Performance is highly optimized. Starting from an annotated stream, `Potamides` infers a ridge-line spline via gradient-based optimization in 10–30 seconds on a 2023 M2 MacBook Pro. Posterior refinement with NUTS (**?**) typically requires another 20 seconds. The potential-inference stage evaluates the curvature likelihood on a dense grid of 10^6 parameter points—marginalizing over roughly 50 spline realizations—in about 15 seconds per stream segment. For a galaxy with three segments, an end-to-end analysis runs in under 15 minutes on standard laptop hardware. On a single GPU, the combined spline-posterior and potential-evaluation stages execute in roughly one minute. This speed supports the rapid evaluation of many potential families, making it easier to test model dependencies without committing early to a single dynamical description.

Research impact statement

`Potamides` is actively used in current research workflows. Specifically, it is being used in a Euclid Key Paper analysis of stellar streams in the Q1 data release and in a separate study of streams in the Stream Legacy Survey. Both projects rely on the package as the primary implementation for curvature-based inference.

The software demonstrates immediate scientific value by providing tested, high-performance, reproducible capabilities. Notably, `Potamides` successfully reproduces the foundational research results of (Nibauer et al., 2023). By streamlining the process from annotating stream segments to calculating potential likelihoods on standard hardware, `Potamides` serves as a highly practical tool for researchers exploring gravitational potentials through stream morphology.

AI usage disclosure

Generative AI tools were used during development to assist with refactoring, drafting, and documentation. They were also used for copy-editing during manuscript preparation. All AI-assisted outputs were reviewed and validated by the human authors, who made all scientific, methodological, and software-design decisions.

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